FOOTBALL PLAYER MARKET VALUE PREDICTION PROJECT REPORT

1.Business Requirement Document (BRD)

1.1 Problem Statement

Football clubs face challenges in accurately estimating player market values, leading to:

- Overpayment or undervaluation in transfers
- Inefficient budget allocation
- Missed opportunities in talent acquisition

The objective is to build a data-driven solution to:

- Predict player market value (continuous variable)
- Identify key factors influencing value
- Evaluate the impact of dimensionality reduction (PCA) and hyperparameter tuning

---

1.2 Business Impact

- Improved accuracy in player valuation
- Better negotiation strategies in transfers
- Enhanced scouting and recruitment decisions
- Optimized financial planning for clubs

1.3 Key Performance Indicators (KPIs)

- Model $R^2 \geq 0.85$
- Reduced Mean Absolute Error (MAE)
- Identification of top contributing features
- Actionable insights for club management

1.4 Success Criteria

- Reliable predictive model
- Explainability of results
- Alignment of insights with football economics
- Usable solution for club decision-making

---

2. Data Understanding Document

2.1 Dataset Overview

The dataset contains player-level information including:

- Demographics (Age, Nationality)
- Performance metrics (Overall rating, Potential)
- Club and position details
- Physical attributes (Height, Weight)

2.2 Target Variable

- **Market Value** (numeric, continuous)

2.3 Feature Classification

Numerical Features:

- weight, height, age, ball _control , dribbling
- marking, slide_tackle, stand_tackle, aggression
- reactions, att_position, interceptions, vision
- composure, crossing, short_pass, long_pass, acceleration

- stamina, strength,balance,sprint_speed,agility
- jumping , heading, shot_power, finishing
- long_shots,curve, fk_acc, penalties, volleys
- gk_positioning, gk_diving, gk_handling, gk_kicking, gk_reflexes

Categorical Features:

- player, country, club

2.4 Data Quality Observations

- Some columns were non-informative (e.g., *marking*, *player*) and were excluded.
- The value column required cleaning to remove currency symbols.
- No null values were present in the dataset.
- Duplicate records were removed using drop_duplicates().
- High-cardinality categorical features (e.g., *Club*, *Nationality*) required encoding. Label Encoding is used.

2.5 Distribution Observations

- Market value distribution is heavily right-skewed
- Log transformation applied to stabilize variance

---

3. Feature Engineering Document

3.1 Feature Selection

Removed:

- player (non-informative)
- marking(non-informative)

3.2 Encoding Strategy

- Label Encoding used for categorical variables

3.3 Feature Scaling

- StandardScaler applied for PCA and linear models

3.4 Dimensionality Reduction

- PCA applied to reduce feature space while retaining ~95% variance

---

3.5 Additional Enhancements

- Log transformation of target variable
- Rare category grouping for categorical features

---

4. Model Development Document

4.1 Models Used

Linear Regression

• Baseline regression model
• Easy to interpret coefficients
• Works well for continuous outcomes with linear relationships

Decision Tree Regressor

• Captures non-linear relationships in data
• Provides interpretability through tree splits
• Prone to overfitting if not tuned

Random Forest Regressor

- Ensemble method combining multiple trees
- Reduces overfitting by averaging predictions
- Provides reliable feature importance rankings

4.2 Model Selection Rationale

- Multiple models used for comparison
- Random Forest selected as final model due to robustness and ability to handle non-linearities

---

5. Model Evaluation Report

5.1 Evaluation Metrics Used

- R^2 (explained variance)
- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)

5.2 Model Comparison (Example Summary)

Model	R2 Score	(MSE / MAE)
Linear Regression	Moderate	High
Decision Tree	High	Lower
Random Forest	Highest	Lowest

5.3 PCA Impact

- Reduced feature redundancy and captured most variance in fewer components.
- PC1 alone explained ~57% of variance, representing attacking technical skills (dribbling, vision, ball control, finishing).
- Models trained on PCA features achieved similar accuracy ($R^2 \approx 0.90$, MSE ≈ 94, MAE ≈ 6.9) compared to original features.
- Improved efficiency and interpretability without sacrificing predictive performance.

5.4 Final Model (With Hyperparameter Tuning)

- GridSearchCV used for Random Forest
- Improved generalization and reduced error

---

6. Insights and Findings

6.1 Key Insights

- A few countries dominate player supply, but global diversity exists
- Clubs are broadly represented, ensuring balanced analysis
- Player careers peak in mid-20s, with limited longevity beyond 30
- Log transformation enhances predictive modeling by reducing bias from extreme values

6.2 Key Drivers of Market Value

- Technical And Attacking Skills
    - Attributes such as ball control, dribbling, short passing, vision, and composure show the strongest correlation with player value. These skills highlight a player's ability to create and execute plays effectively.

- Cognitive Attributes

    - Reactions and attacking positioning are highly influential, reflecting a player's awareness, decision-making, and ability to adapt during matches

- Offensive Capabilities
    - Shooting-related features (finishing, shot power, long shots, volleys, free-kick accuracy) contribute significantly, emphasizing the importance of scoring ability in market valuation

- Age
    - Most players are in their mid-20s, fewer remain active beyond 30, and careers rarely extend past 35.

- Physical Attributes
    - While traits like stamina, sprint speed, acceleration, and strength play a role, their correlation with value is moderate compared to technical and cognitive skills

- Club & Country correlations are weak individually, but they matter indirectly (elite clubs/nations tend to have higher-value players)

---

7. Business Recommendations

7.1 Transfer Strategy

- Use model predictions to benchmark against market prices
- Avoid overpaying for players with inflated values

7.2 Scouting

- Reactions are a critical technical attribute influencing valuation — clubs should prioritize scouting players with strong reaction skills, as this trait directly boosts market worth.
- Identify undervalued players with high potential
- Focus on attributes most predictive of future value

7.3 Financial Planning

- Optimize squad composition based on value-to-performance ratio

7.4 Player Development

- Use model insights to design training programs that strengthen moderately correlated attributes (e.g., stamina, positioning, shot power).
- Track progress with predictive models to measure how improvements

---

8. Final Conclusion

Successfully built a machine learning model to predict football player market value.

- Identified key factors influencing value
- Evaluated PCA and hyperparameter tuning impacts
- Provided actionable recommendations aligned with club goals

---

9. Evaluation Criteria Mapping

Criteria	Addressed
Problem Understanding	Yes
Model Performance	Yes
Feature Engineering	Yes
Insights Quality	Yes
Documentation	Yes